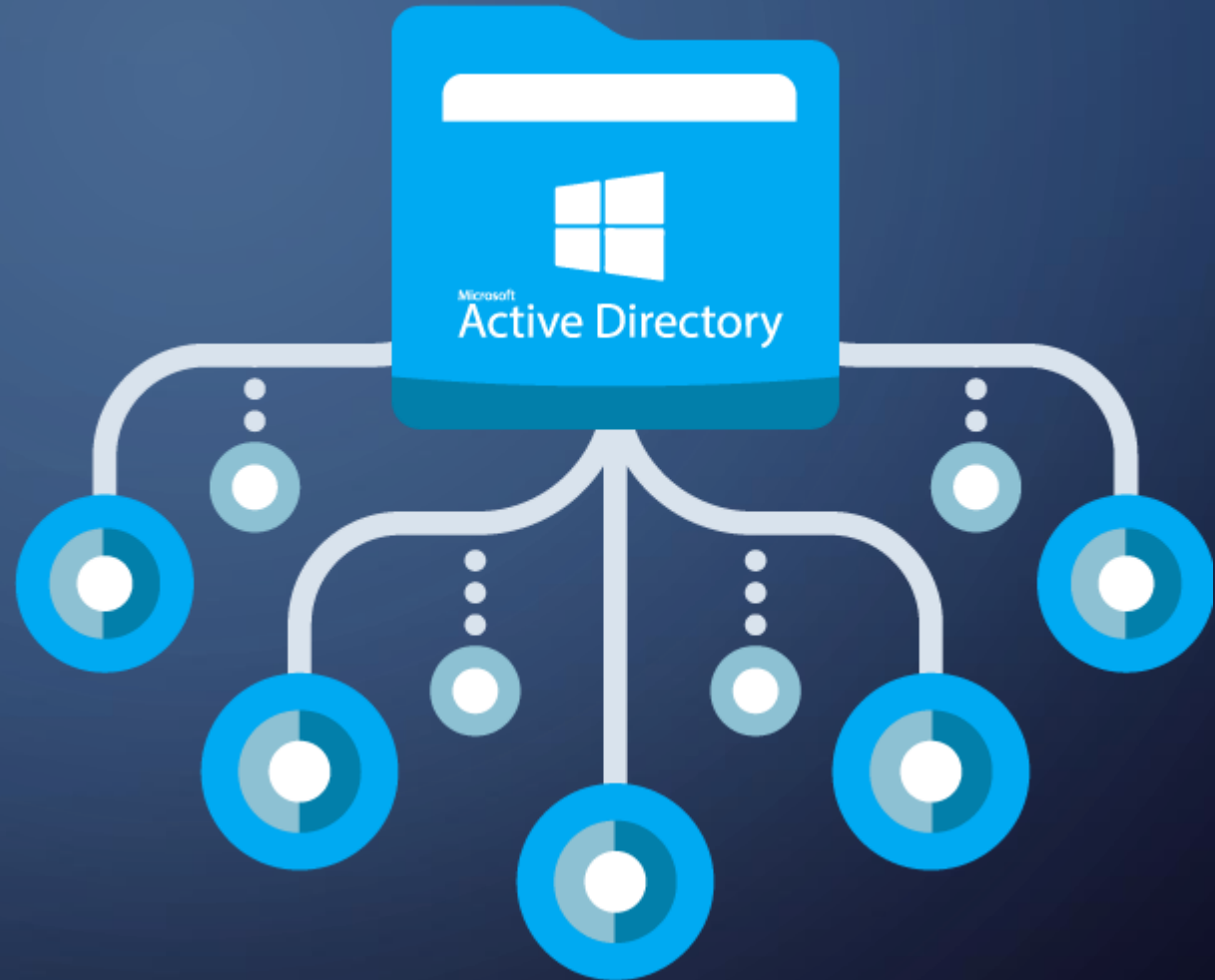


CREATING AN ACTIVE DIRECTORY

Hashan Kodippilige
Network Security - Project 1

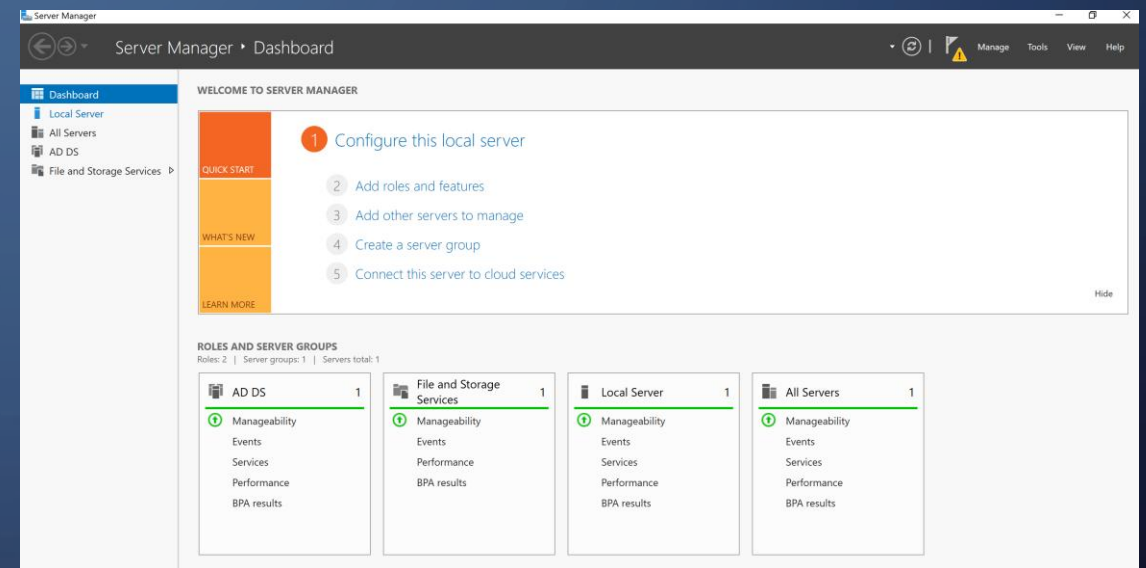
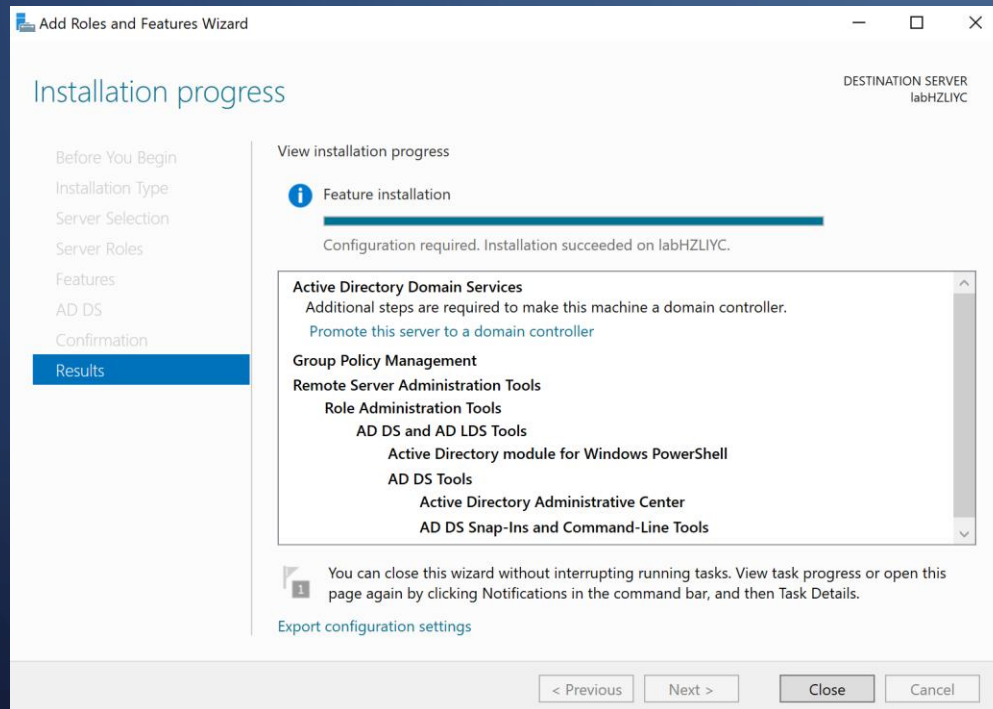


INTRODUCTION

- Active Directory (AD) is a directory service developed by Microsoft.
- AD plays a vital role in cybers operations because of its centralized control structure.
- AD manages authentication, authorization, and policy enforcement across an organization .
- AD verifies user identities and determines what resources they are permitted to access.
- Organizational cybersecurity teams rely on AD to create, modify, disable, and monitor user accounts.

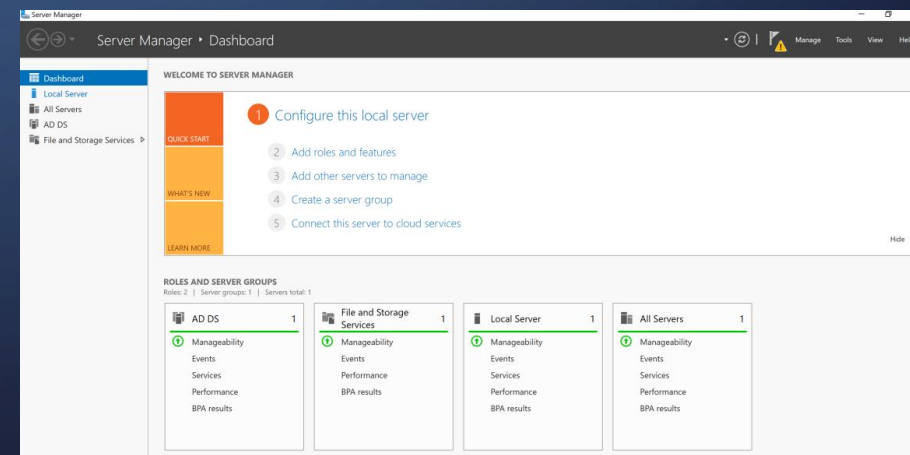
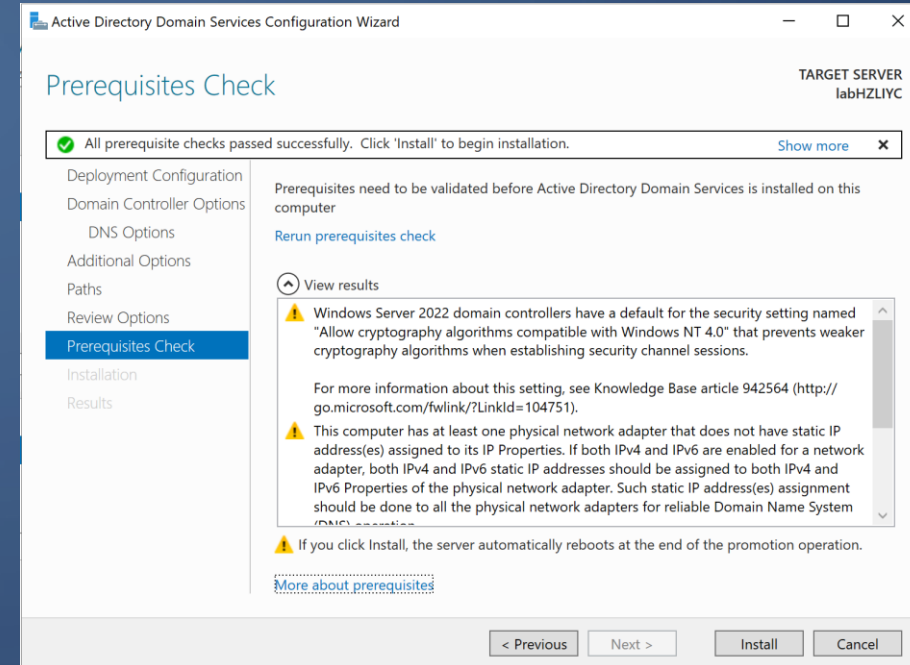
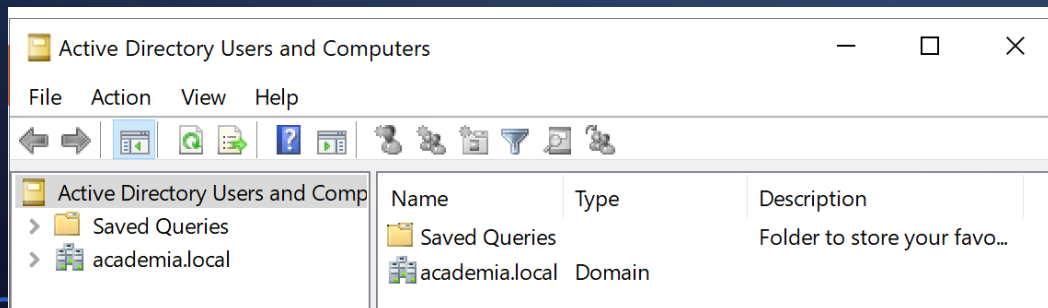
INSTALLING ACTIVE DIRECTORY

- Using 'Add Roles and Features', installed Active Directory.



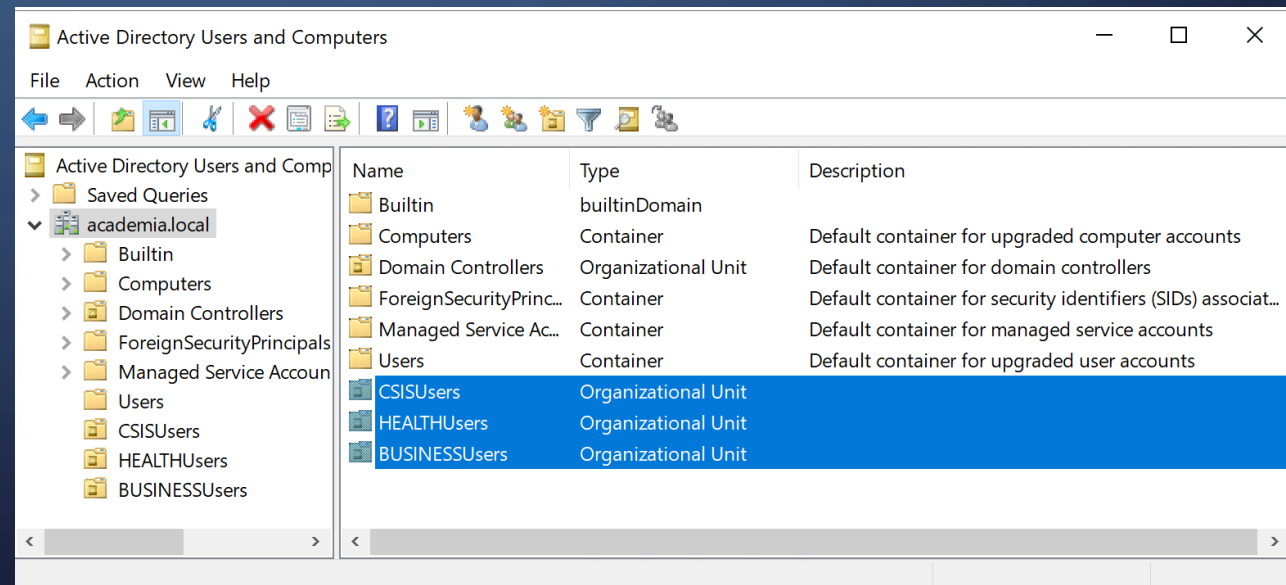
DOMAIN CONTROLLER

- Domain Name: academia.local
- Server: Windows Server 2022
- Role Installed: Active Directory Domain Services
- Promoted Active Directory to a Domain Controller
- DNS configured automatically



LAUNCHING ACTIVE DIRECTORY USERS AND COMPUTERS

- Created three Organizational Units (OUs) for
 - CSISUsers
 - HEALTHUsers
 - BUSINESSUsers
- OUs allow logical grouping of users and help enforce security policies.



CREATING USERS

- Created users under following Organizations as follows;
 - CSISUsers - Hashan Kodippilige
 - BUSINESSUsers - Jack Smith
 - HEALTHUsers – Nick Powell
- Password policy enforced: User must change password at next logon

New Object - User

Create in: academia.local/CSISUsers

When you click Finish, the following object will be created:

Full name: Hashan Kodippilige
User logon name: HKodippilige@academia.local
The user must change the password at next logon.

< Back Finish Cancel

New Object - User

Create in: academia.local/BUSINESSUsers

When you click Finish, the following object will be created:

Full name: Jack Smith
User logon name: JSmith@academia.local
The user must change the password at next logon.

< Back Finish Cancel

New Object - User

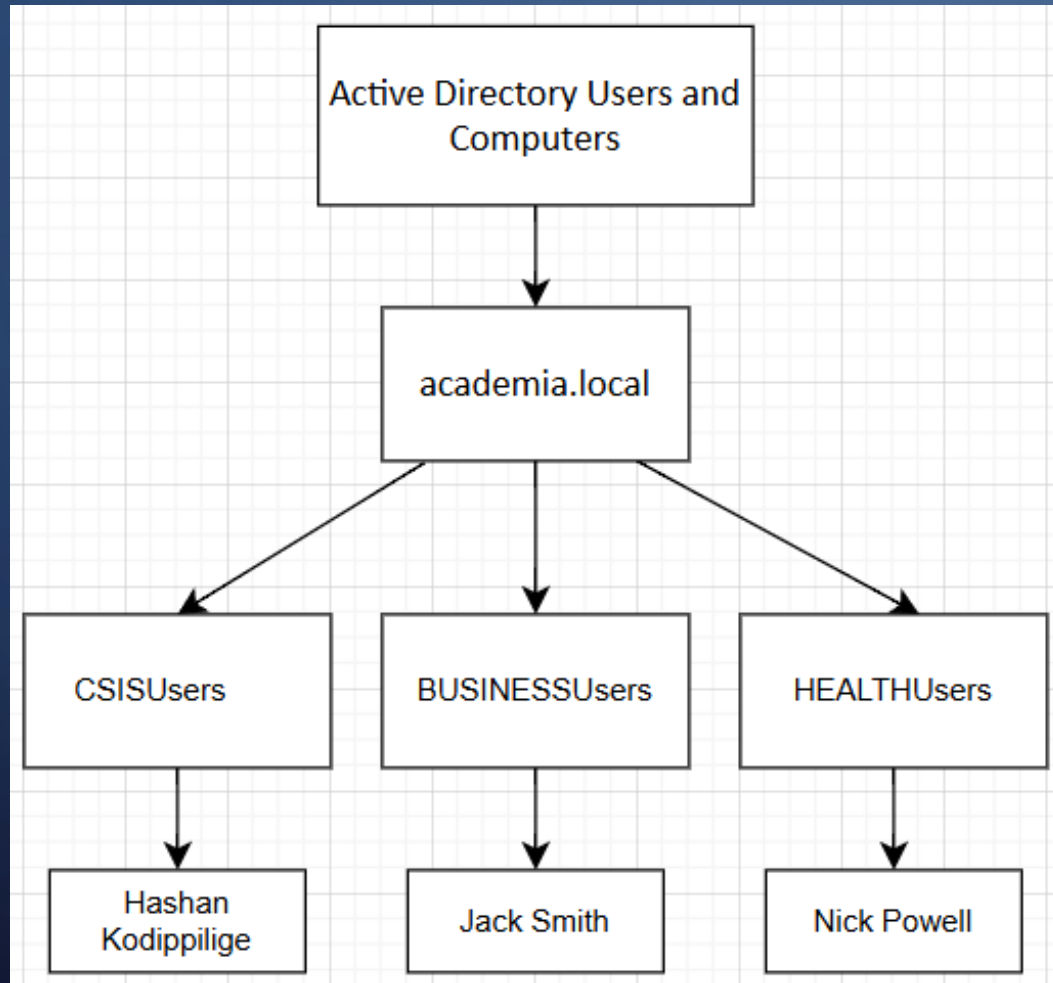
Create in: academia.local/HEALTHUsers

When you click Finish, the following object will be created:

Full name: Nick Powell
User logon name: NPowell@academia.local
The user must change the password at next logon.

< Back Finish Cancel

ORGANIZATIONAL STRUCTURE



RISKS POSED BY THE ACTIVE DIRECTORY TOOL

- Centralized control makes it a high-value target.
- Stores sensitive authentication credentials.
- Misconfigurations can expose excessive privileges.
- Weak password policies increase attack success.
- Single successful attack can result in widespread system compromise, data breaches, and operational disruption.

HOW AD TOOL CAN BE EXPLOITED BY HACKERS

- Credential-Based Attacks
 - Pass-the-Hash: Reuse stolen NTLM hashes without knowing passwords
 - Credential Dumping: Extract passwords from memory using tools like Mimikatz
- Privilege Escalation
 - Exploit misconfigured permissions
 - Escalate to Domain Admin access
- Group Policy Abuse
 - Modify GPOs to deploy malware
 - Disable security settings domain-wide

CONCLUSION

- Centralized identity management system was established by promoting the server to a Domain Controller and creating the domain *academia.local*.
- Active Directory allows logical structuring of users and resources within an enterprise.
- AD security plays a central role in authentication, authorization, and the overall security of a Windows-based environment.
- Dividing departments highlights the importance of least-privilege access control.
- Misconfigured or poorly secured Domain Controller would lead to widespread system access, data breaches, and operational disruption.

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<https://www.sophos.com/en-us/cybersecurity-explained/active-directory-security>

The background is a dark blue gradient. In the corners, there are white line-art illustrations of circuit boards or neural networks, with lines and small circles representing nodes.

THANK YOU!